STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 9D

Description

9.**Write R program to find the following Regressions with the sample data and visualize the regressions graphically.**

d) Poisson Regression.

Procedure and Overvation:

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Poisson regression involves regression models in which the response variable is in the form of count and not fractional number.

for example, the count of number of births or number of wins in a football match series.

Also the valyesm of response variable follow a piosson distribution.

the general equation for piosson regression if log y = a+b1x1+b2x2+....+bnxn)

y is response variable

a,b1,b2,...bn are coefficients

x1, x2, ...xn are predictor variables

the fucntion used to create the regression model is the glm()

syntax :

glm() in logestic regression is

glm(formula, data, family)

formula is the symbol presenting the relationship between the variables

data is the dataset giving the values of those variables

family is R object to specify the details of the model,its value is binomial for logistic regression

SOURCS CODE

```
input<-warpbreaks

print(head(input))

output<-glm(formula = breaks ~ wool+tension,data=warpbreaks,family = poisson)

print(summary(output))
```

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Console Jobs

R 4.2.0 · ~/

```
> output<-glm(formula = breaks ~ wool+tension,data=warpbreaks,family = poisson)
> source("~/.active-rstudio-document")
  breaks wool tension
1     26    A       L
2     30    A       L
3     54    A       L
4     25    A       L
5     70    A       L
6     52    A       L

Call:
glm(formula = breaks ~ wool + tension, family = poisson, data = warpbreaks)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-3.6871  -1.6503  -0.4269   1.1902   4.2616

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.69196    0.04541  81.302  < 2e-16 ***
woolB       -0.20599    0.05157  -3.994 6.49e-05 ***
tensionM    -0.32132    0.06027  -5.332 9.73e-08 ***
tensionH    -0.51849    0.06396  -8.107 5.21e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 297.37  on 53  degrees of freedom
Residual deviance: 210.39  on 50  degrees of freedom
AIC: 493.06

Number of Fisher Scoring iterations: 4

>
```

VIVA RESPONSES

Q) Linear regression analysis is used to predict the value of a variable based on the value of another variable.

True

Q) A Linear Regression model's main aim is to find the best fit linear line and the ___ of intercept and coefficients such that the error is minimized.

Optimal values

Q) Error is the difference between the actual value and Predicted value and the goal is to reduce this difference.

True

Q) The process of quantifying data is referred to as ___.

Enumeration

Q) Text Analytics, also referred to as Text Mining?

True